Feedback Loops

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach feedback using explicit event logs and denominators.

Supply complete scripts, preserve implementation review and assess the current exposure-log task consistently.

Keep one habit: Require a policy/exposure/feedback chain and a limit on any preference claim.

MAIN LESSON · SECTION 1

Prepare the feedback-loop investigation

> Keep current goals and assessment separate from earlier implementation review.

Current targets and preparation

Main targets are exposure, clicks and selection feedback. Prepare the two scripted three-round logs and a fresh X/Y/Z independent audit. Use invented events; no real user testing or personal profiles are needed. The source case-exposure and implementation notes below concern retained scoring-function review, not the new feedback-loop assessment.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Retrieve prior score/constraint knowledge

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a weighted score

Reference example, not the only acceptable wording: Score 8.

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

Look for: Represent items with numeric features

Reference example, not the only acceptable wording: Energy 4, group work 1.

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Look for: Detect absent and unusable values

Reference example, not the only acceptable wording: 0 is a recorded allowed duration. The other two are unknown under this schema.

No. Filter by the hard constraint before ranking.

Flexible teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

▼ Review · Earlier implementation setup

Primary target: Implement a **deterministic recommender**

Case **exposure** check: this source example appears in ai-2/24, ai-2/33. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Teach Week 24 as a two-session investigation using fictional data. Build a fictional activity **recommender** using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name.

This makes the outcome reproducible rather than dependent on **input** order. Assume unique names and validated numeric fields. Assess the three stated outcomes through the independent task, not by copying the worked result.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Recommendation Scores

Similarity by Selected Traits

Missing Data

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A **recommender** shows poetry in 2 slots and adventure in 20. Poetry gets 1 click; adventure gets 6. Compute clicks per shown slot and explain why neither rate alone proves preference.

Concept to explain

Recommendations influence behavior, which then becomes new data.

Term	Meaning
feedback loop	A cycle in which outputs affect later inputs and therefore future outputs.
exposure	The opportunity to see or encounter an item, which can affect later choices.
interaction data	Records of what people were shown and how they acted, such as views, clicks or corrections.
narrowing	Reducing the allowed choices, scope or interpretation of a task.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; **explanation** 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

By the end, I can explain how recommendation **exposure** can create a **feedback loop**.

MAIN LESSON · SECTION 2

Teach the exposure-feedback mechanism before interpreting a click total

> Keep the retained implementation program explicitly separate.

State the concrete feedback loop

A **recommender** shows only adventure books, uses clicks to learn and then shows more adventure.

Explain the **feedback loop**.

Exposure affects clicks; clicks affect future **exposure**. Record what was shown and offer controlled variety.

Trace the pictured initial tie and3rounds

A fictional catalogue repeatedly recommends two books using observed clicks.

Initially each click count is 0. Equal counts choose earlier ID A over B.

Only the recommended book is shown. In an invented three-round record, A is shown and clicked each round; B is never shown.



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the observation and preference limits

The selection policy changes which **evidence** is available. A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books.

Keep display counts separate from click counts; investigate alternatives with a teacher-approved **exposure** plan rather than treating zero as negative feedback.

A click can reflect curiosity or a mistake; even equal **exposure** does not reveal every reason for selection.

Predict the one-display nonclick trial

Change: On a planned trial, show B once and record no click.

Prediction: B gains one **exposure** but still zero clicks.

Reason: The **observation** now supports a nonclick on that display, not a permanent dislike.

Run the complete scripted comparison

```
def simulate(plan=None):
    exposures = {"A": 0, "B": 0}
    clicks = {"A": 0, "B": 0}
    scripted_click = {"A": True, "B": False}
    log = []
    for round_number in range(3):
        if plan is None:
            shown = sorted(clicks, key=lambda name: (-clicks[name], name))[0]
        else:
            shown = plan[round_number]
        exposures[shown] += 1
        clicked = scripted_click[shown]
        if clicked:
            clicks[shown] += 1
        log.append((round_number + 1, shown, clicked, exposures.copy(),
clicks.copy()))
    return log, exposures, clicks
for label, plan in [("click-count policy", None), ("planned exposure", ["A", "B",
"A"])]:
    log, exposures, clicks = simulate(plan)
    print(label)
    for event in log:
        print(event)
    print("displays", exposures, "clicks", clicks)
```

Elicit the exposure-field explanation

Use the diagram and verbal **explanation** together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Review · Earlier scoring-function explanation

Build a fictional activity **recommender** using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name.

This makes the outcome reproducible rather than dependent on **input** order. Assume unique names and validated numeric fields.

The function prints nothing; the caller prints A. best starts as None to mean no eligible option found. Python short-circuits or: when best is None, later comparisons in that condition need not run.

After selecting an item, best_score is numeric. An empty list or no eligible item returns None.

> **Predict the result if the list order is reversed, and if the limit is 8.**

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. System recommends mostly fantasy.

2. Reader can only click among shown books.

3. Clicks are treated as stronger fantasy preference.

4. Next round shows even more fantasy, hiding other possible interests.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Term	Meaning
Recommender	A procedure that suggests options using a stated policy
Tie	Two options have the same comparison score
Deterministic	The same inputs and rules produce the same result
Explanation	Evidence describing how a result was obtained

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Feedback Loops

The facts we will use

AI ACADEMY / AI-2/24

Exposure and clicks after the invented three rounds

	Shown	Clicked
A	3	3
B	0	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/24

Feedback Loops

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial tie

Counts A=0,B=0; ID rule shows A.



2. Record event

A receives a click, so its count becomes 1; B stays 0.



3. Repeat selection

Higher A count causes A to be shown again; after three clicks A=3,B=0.



4. Interpret carefully

B's zero may mean no exposure, not dislike.

CHANGE ONE THING

On a planned trial, show B once and record no click.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Initial tie	Counts A=0,B=0; ID rule shows A.
2. Record event	A receives a click, so its count becomes 1; B stays 0.
3. Repeat selection	Higher A count causes A to be shown again; after three clicks A=3,B=0.
4. Interpret carefully	B's zero may mean no exposure, not dislike.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Guide the current event-log reasoning, then fade help

> Collect individual reasoning before revealing the key.

Main sequence

Main sequence: state exposure/click meanings and scripted events; predict the click-count log; trace the pictured loop; compare the predeclared A,B,A plan; complete current W1–W3 with fading support; collect new X/Y/Z W4 before feedback; use current W5 to limit the diversity claim. The earlier deterministic-function schedule is optional review.

Current completion task and key

Completion task: B was never shown, then a planned trial shows B once with no click. Give the before/after exposure and click counts, identify which click rate is unavailable and limit the inference.

Teacher check: Before:0 displays,0 clicks, rate unavailable. After:1 display,0 clicks, observed rate0/1=0%; this is one observed nonclick, not a permanent dislike.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Current misconception check

Proposed **explanation**: Zero clicks proves dislike even when the item was never displayed. Use the exposure denominator to repair the claim.

Expected correction: No exposure means no observed click rate or opportunity; it is not measured dislike.

Collect **explanation**s, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group **explanation** while others solve a more demanding variation.

Pictorial exposure question

What extra field would reveal why comparing raw clicks is misleading?

Display or **exposure** count: A shown three times, B zero. Raw clicks confound opportunities with response.

Changed-case reasoning: B gains one **exposure** but still zero clicks. The **observation** now supports a nonclick on that display, not a permanent dislike.

Collect each learner's **explanation** before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep every learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an **explanation**.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Review · Earlier implementation sequence

Reversing still returns A. Limit 8 leaves no eligible item and returns None. Exit: X meets the 12-minute limit and ties for highest eligible score 9; the alphabetical tie rule selects X.

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.

Stage	Teacher action
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: A **recommender** shows only adventure books, uses clicks to learn and then shows more adventure.

Write the question to explain: Explain the **feedback loop**.

Reveal after **prediction**s: **Exposure** affects clicks; clicks affect future **exposure**. Record what was shown and offer controlled variety.

Use the earlier changed question as optional later practice. Reserve Audit a new exposure log independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Case-specific teaching sequence

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied event-log comparison

> State the scripted response assumptions beside any result.

Current investigation and answer guidance

Scripted click-count policy showsA,A,A: displaysA3/B0,clicksA3/B0. PlannedA,B,A gives displaysA2/B1,clicksA2/B0; displayed-ID diversity rises1→2 only in this simulation. Both runs use the stated A-click/B-nonclick events; they do not measure real enjoyment. B0/0 is unavailable before exposure; after one supplied nonclick B0/1 is observed0%, not permanent dislike. Reserve the fresh X/Y/Z audit for current independent W4.

Predict, simulate and inspect

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Use the supplied three-round click-count policy and A,B,A exposure plan under the same scripted events. Log displays and clicks separately and compare distinct displayed IDs. Do not claim real preference or satisfaction was measured.

4. What did the hardest test teach you that the normal test did not?

Keep event evidence

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Current misconception support

Current misconceptions: zero clicks equals dislike—check whether any display occurred; click share measures enjoyment—clicks can reflect curiosity, position or error; two displayed IDs proves a better recommender—it measures only this simulation's exposure diversity; a planned trial guarantees an outcome—retain unobserved responses as unknown. Keep the X/Y/Z independent key out of shared practice.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral **explanation**, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Review · Earlier implementation misconceptions

A successful run proves the conclusion: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action.

Display the chosen rule and let a person review whether an option is suitable.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an **explanation**. Fresh retry: B (2,4,8), C (3,2,8), A (5,5,9), limit 8. B/C tie at 8; B wins. Limit 7 returns None.

A convenient shortcut preserves meaning: Updating best on score >= best_score makes the last tied item win, so reordering can change results. Use the explicit name comparison for ties.

Test both orders and a no-match case after the repair.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Audit a new exposure log independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the current X/Y/Z exposure-log transfer

> Reveal this key only after the first independent attempt.

Current independent question and checked key

A fresh fictional log has X displayed4 times with1 click, Y displayed4 times with2 clicks, and Z displayed0 times with0 clicks. Calculate the observed click rates for X and Y; state why Z has no measured click rate. Explain how always selecting the highest raw-click item can shape future evidence. Then a planned trial shows Z once with no click: update Z's counts and rate, and state one limit of that observation.

X has1/4=25% and Y2/4=50% observed click rates. Z has no exposure and0/0 is unavailable, not measured0%. Selecting Y repeatedly can increase Y's opportunities while leaving Z unobserved. After the supplied Z display with no click, Z has1 display,0 clicks and observed0/1=0%; one nonclick does not establish permanent dislike or future performance.

Judge reasoning and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Current workbook answer guidance

Current W1: round1 A1/B0 displays and clicksA1/B0; round2 A2/B0 andA2/B0; round3 A3/B0 andA3/B0. Current W2: beforeB0/0 unavailable; afterB0/1=0% on one display. W3: raw counts reflect unequal opportunities; propose a documented exposure plan without claiming its unknown outcomes. W5: distinct displays1 versus2 is a simulated count, not enjoyment or educational evidence.

▼ Review · Earlier implementation questions and keys

W1 Trace selection

Why does A win the worked example even though B is **process**ed first?

Both score 8 and qualify; A is alphabetically first under the tie rule.

W2 Handle no match

State recommend([], 10) and the worked items with limit 8.

Both return None; the caller should display no eligible option.

For the empty list, the loop has no iterations, so best stays None. With the worked durations 10 and 9 under a limit of 8, neither item passes the eligibility check, so best again stays None.

The caller must turn this result into a clear no-eligible-option message; it must not treat None as an activity name.

W3 Repair order dependence

A tied selection changes after reversing **input** order. Identify the likely missing policy and test.

An explicit **deterministic** tie rule is missing or incorrect. Apply alphabetical name order and rerun both orders.

W4 Implement independently

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

X and Y score 9; Z is excluded. X wins alphabetically in either order. Limit 9 returns None.

W5 Explain the result

Write a one-sentence **explanation** of X winning without calling it objectively best.

X meets the 12-minute limit and ties for highest eligible score 9; the alphabetical tie rule selects X.

Assess independent transfer

Selected case: Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

Selected case answer

X and Y score 9; Z is excluded. X wins alphabetically in either order. Limit 9 returns None.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Exposure affects clicks; clicks affect future **exposure**. Record what was shown and offer controlled variety.

Additional transfer practice question

Only space books are shown, so nearly every click is on space. Can these clicks prove readers dislike other topics? Explain the loop.

Reasoned practice answer

No. **Exposure** affects what can be clicked. Showing space produces space clicks that may cause even more space recommendations; test other relevant options to learn more.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Clicks reveal preferences independently of what is offered.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Poetry is 1/2=50%; adventure is 6/20=30%. Raw clicks favour adventure but **exposure** rates differ. Position, user mix and tiny poetry sample confound interpretation. Run a balanced comparison and record **exposure**s and outcomes.

Explanations for the additional workbook examples

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Reserve some varied suggestions and distinguish ‘not shown’ from ‘not liked.’

Keep the decision, **input**, result, **explanation** and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Log **recommender exposure**, choice and explicit feedback separately; consider whether its own policy narrows future **evidence**.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Use separate display and click counters with one card per event. A never-shown card stays outside the rate calculation. Extension: compare two predeclared exposure plans under the same scripted event assumption and state why a real causal claim would need more evidence.

For the current independent case require X1/4=25%,Y2/4=50%,Z rate unavailable at0 displays, a policy→exposure→feedback **explanation**, and after the supplied trial Z0/1=0% without a permanent-dislike claim. Fresh retry: P2 displays/1click,Q0/0,R4/1; require

denominators and a documented future **evidence** plan before feedback.

Plan the delayed check

Suggested return points in this level: ai-2/25, ai-2/27, ai-2/30. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 25: Text as Pieces. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week25: Text as Pieces. Carry forward the distinction between observed records and inferred meaning when choosing a text representation.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Implement a **deterministic recommender**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: State recommend([], 10) and the worked items with limit 8.

Reasoned practice answer: Both return None; the caller should display no eligible option.

Check every learner before discussion: Use the worked case for Implementing a **recommender**. Proposed **explanation**: “A successful run proves the conclusion”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action.

Display the chosen rule and let a person review whether an option is suitable.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken **explanation** or drawing; retain the reasoning goal.

Extension: On a planned trial, show B once and record no click. Require a **prediction** and a reason before comparing. Expected result: B gains one **exposure** but still zero clicks.

Reason: The **observation** now supports a nonclick on that display, not a permanent dislike.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's **explanation** and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-**explanation** links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess **explanation**, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an **explanation**. The branch-boundary and indexing lessons deliberately include failure cases.

▼ Reference: teach with the six pictures

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Feedback Loops: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

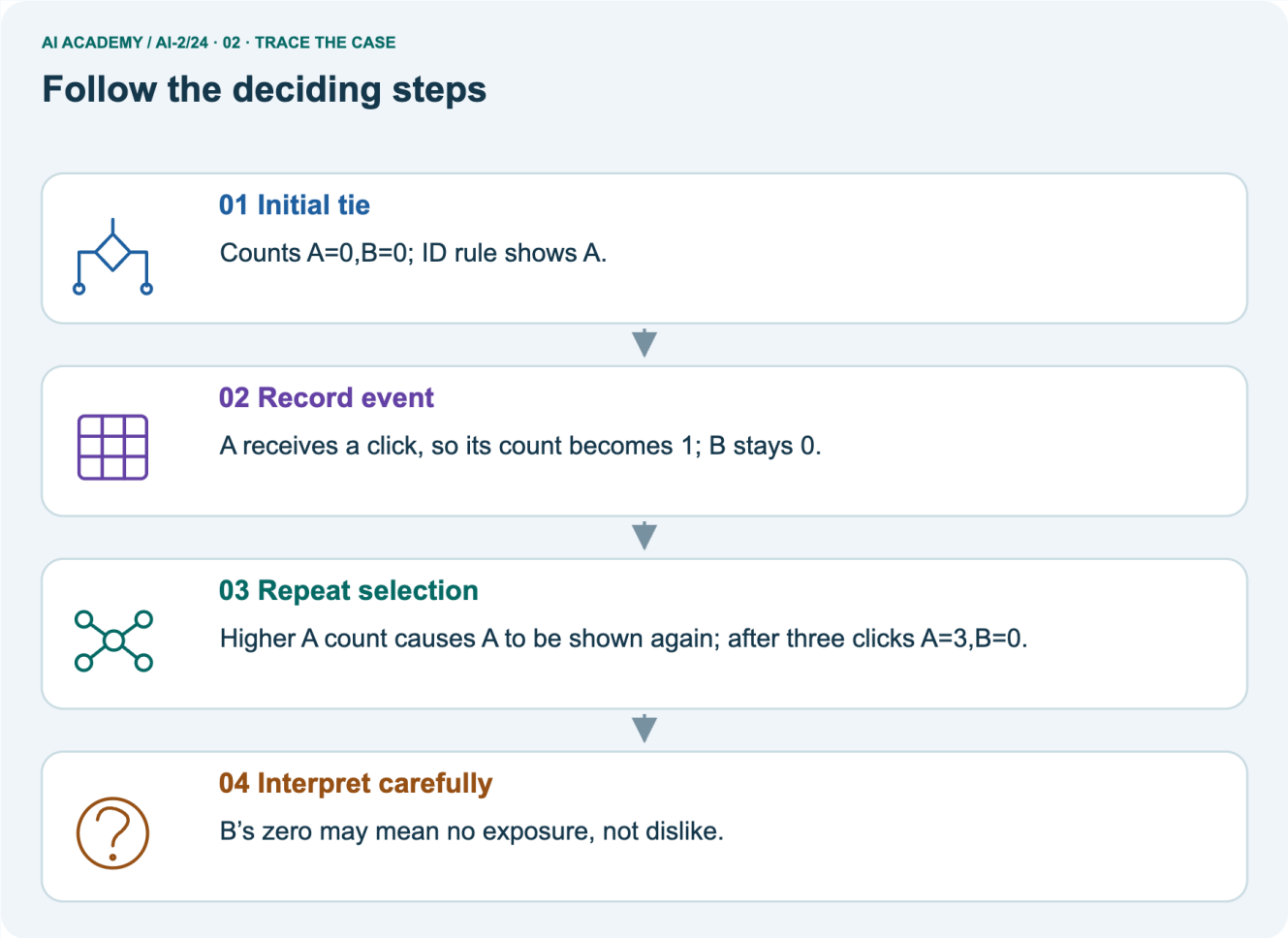
A fictional catalogue repeatedly recommends two books using observed clicks. Supplied facts: Initially each click count is 0. Equal counts choose earlier ID A over B. Only the recommended book is shown.

In an invented three-round record, A is shown and clicked each round; B is never shown.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



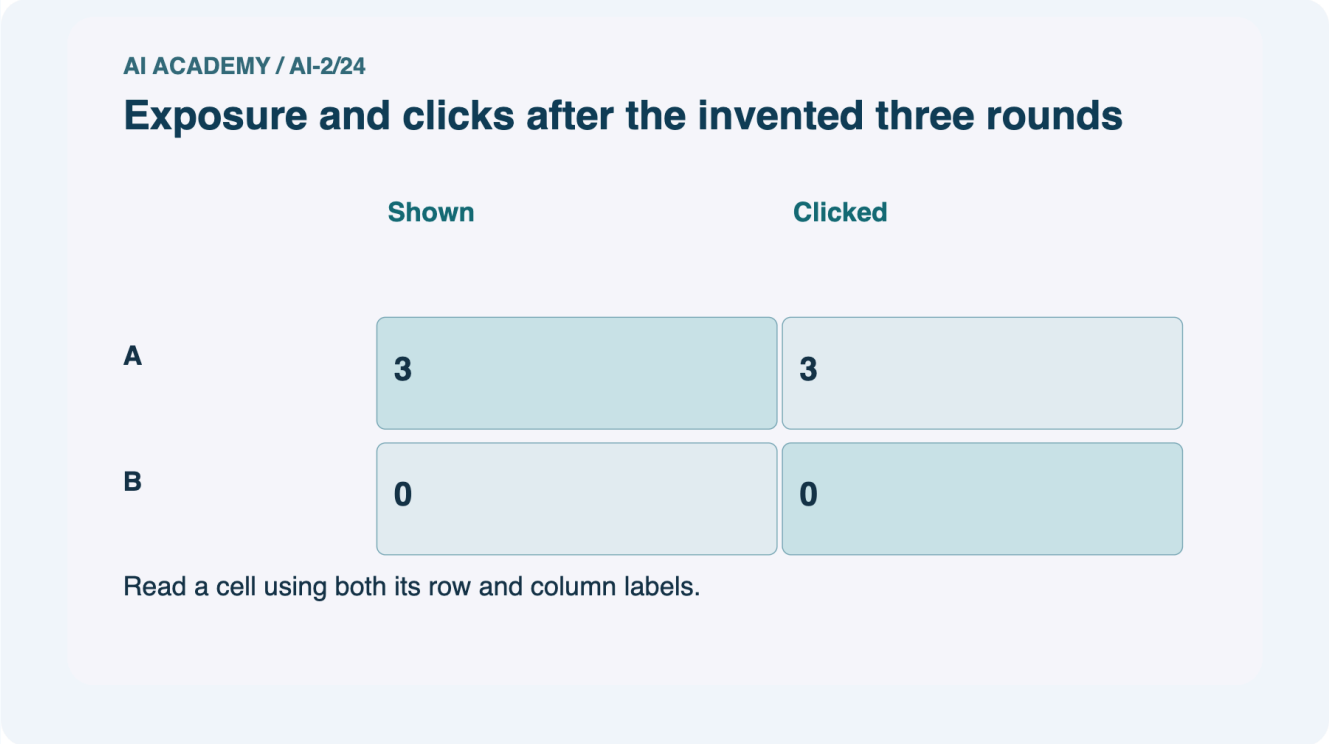
Follow the labelled picture with the matching explanation. Open the vector diagram

Initial tie: Counts A=0,B=0; ID rule shows A. → Record event: A receives a click, so its count becomes 1; B stays 0. → Repeat selection: Higher A count causes A to be shown again; after three clicks A=3,B=0. → Interpret carefully: B's zero may mean no **exposure**, not dislike.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Exposure and clicks after the invented three rounds



Follow the labelled picture with the matching explanation. Open the vector diagram

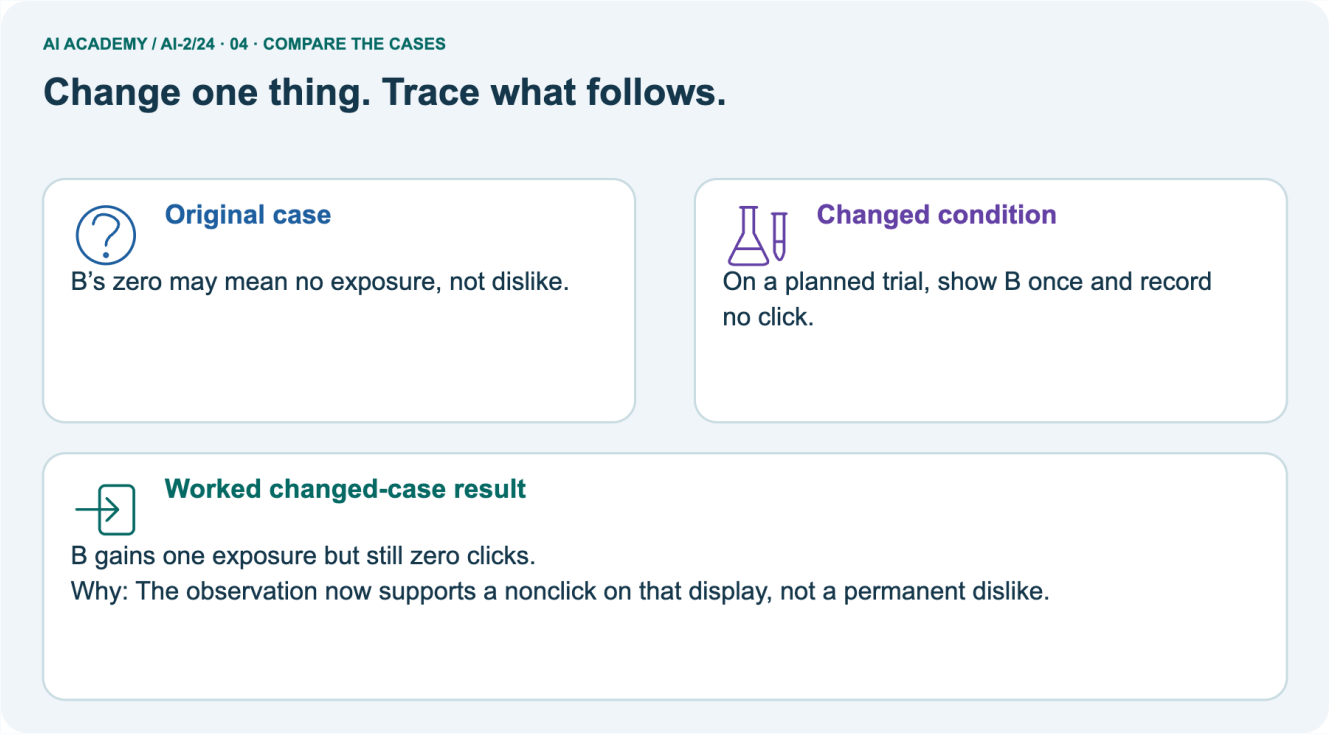
Exposure and clicks after the invented three rounds. A: Shown = 3, Clicked = 3. B: Shown = 0, Clicked = 0.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B's zero may mean no **exposure**, not dislike. Changed condition: On a planned trial, show B once and record no click. Expected result: B gains one **exposure** but still zero clicks.

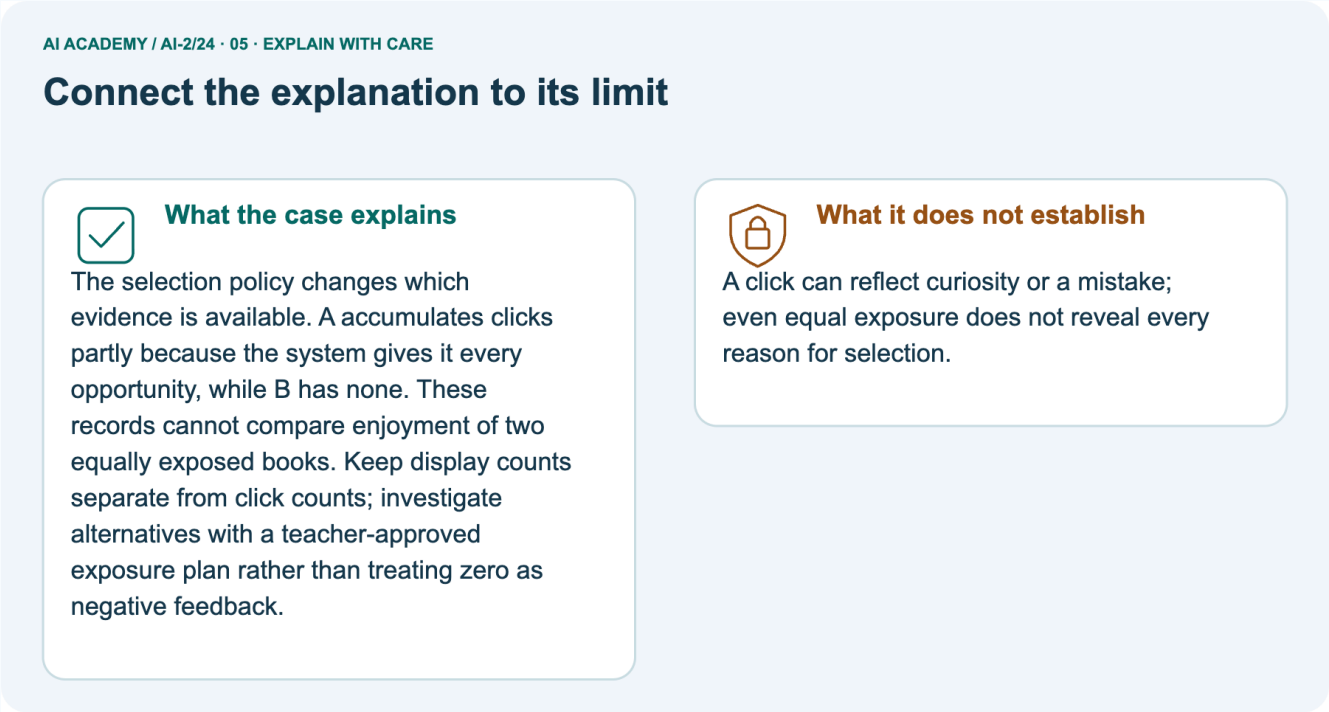
Reason: The **observation** now supports a nonclick on that display, not a permanent dislike.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The **observation** now supports a nonclick on that display, not a permanent dislike.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The selection policy changes which **evidence** is available. A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books.

Keep display counts separate from click counts; investigate alternatives with a teacher-approved **exposure** plan rather than treating zero as negative feedback.

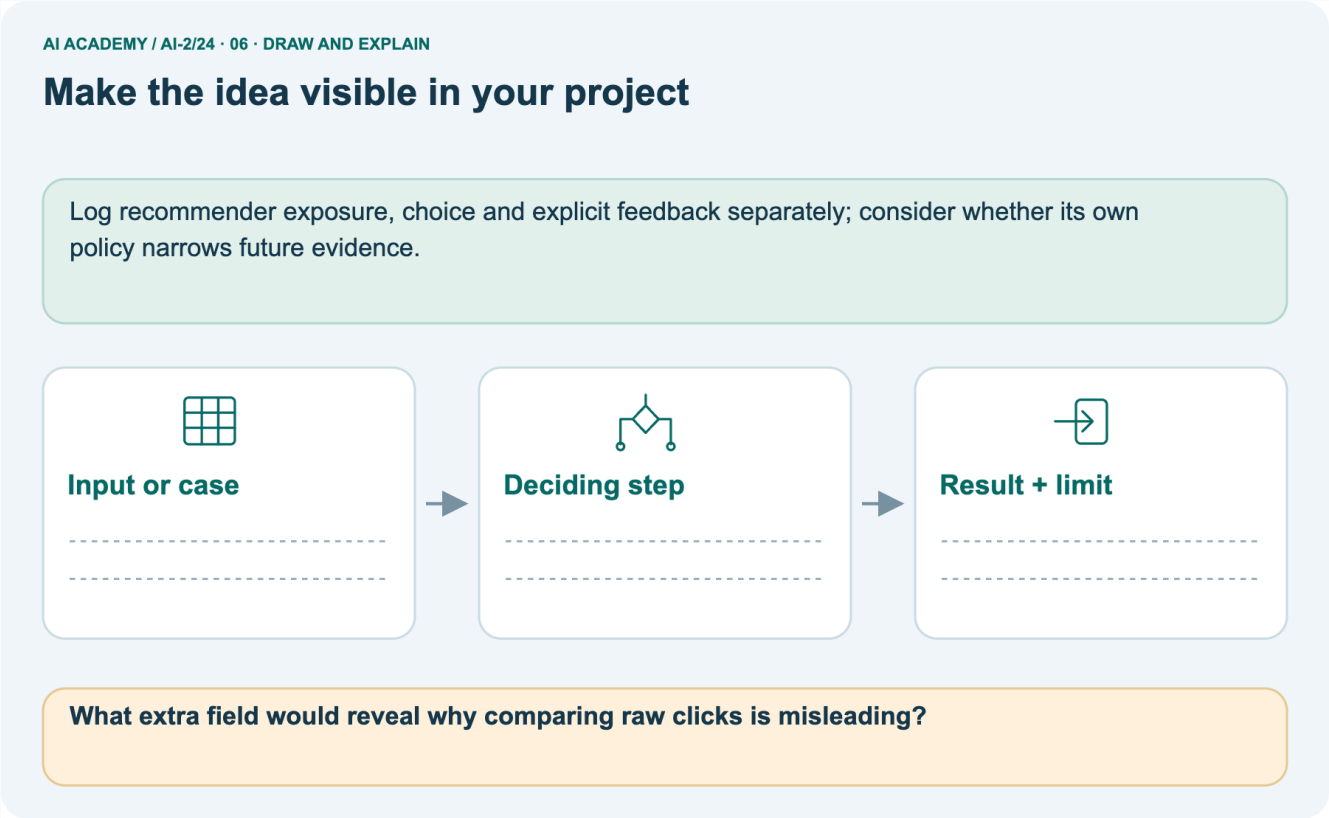
Limit: A click can reflect curiosity or a mistake; even equal **exposure** does not reveal every reason for selection.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A click can reflect curiosity or a mistake; even equal **exposure** does not reveal every reason for selection.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Log **recommender exposure**, choice and explicit feedback separately; consider whether its own policy narrows future **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: What extra field would reveal why comparing raw clicks is misleading?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Display or **exposure** count: A shown three times, B zero. Raw clicks confound opportunities with response.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Feedback Loops: the case

02. Follow the deciding steps

03. Exposure and clicks after the invented three rounds

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.